

LCA at Tallgrass: One-Page Brief

Oct 2, 2026 · Andrew Gibson

How an LCA works

An LCA counts every greenhouse gas emitted and removed over a product's life. For Trailblazer, it sets each shipper's 45Z credit, its clean fuel credits and its removal-credit yield. Four phases (ISO 14040/44):

1. **Goal and scope:** what unit, which boundaries (corn field to Wyoming storage).
2. **Inventory:** every input and output, from fertilizer to booster compression power.
3. **Impact:** add it up in CO₂e, minus land-use and supply-chain charges.
4. **Interpretation:** test what drives the answer (land use, grid power, leakage).

One tonne, four rulebooks

Program	Unit	Land-use charge (iLUC)	What CCS earns
45Z	g/MJ fuel	Excluded after 2025	About 30 CI points; \$195–260/t to 2029
CA LCFS	g/MJ fuel	Included	CI cut, with a CARB CCS pathway
45Q	Tonne stored	Not counted	\$85/t, metered
Puro / Microsoft	Net tonne removed	Charged (~1.5 kg/gal)	Net credits of ~40–50% of captured tonnes

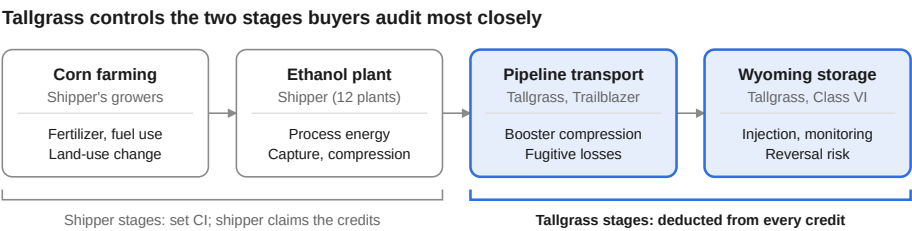
iLUC is the carbon released when growing corn for fuel pushes farming onto new land elsewhere. It is the single biggest swing: Puro's charge alone halves net credits per gallon.

The market in numbers

Market	Size	Latest price	Trend	Trailblazer access
45Z	~\$12.5B/yr federal cost	\$195–260/t CO ₂ (est.)	Ends Dec 31, 2029	Yes, via shippers
45Q	~\$1B/yr, rising	\$85/t	Fixed; indexed from 2027	Yes; the floor after 2029
Clean fuel (CA, OR, WA)	79M CA credits traded in 2025	CA \$77, OR \$147, WA \$76 (Aug 2026)	CA fell \$192 (2019) to \$60 (2025), now recovering	Needs a certified CCS pathway
Durable removals	30.4 Mt contracted in 2025, ~\$8–10B (est.), 7x 2023	BECCS avg \$250–330; Trailblazer likely \$100–200	Avg fell \$490 to \$320 (2023–24)	Yes; ADM Columbus certifying 800+ kt/yr
Voluntary, all types	~\$1.4B primary sales, 2025	Mostly cheap avoidance credits	90% of retirements are avoidance	Indirect
Cap-and-trade	~\$6.7B/yr state proceeds	\$32–40 per allowance	Rising	None for Wyoming storage

Two numbers to remember: Microsoft holds 78.5% of contracted removal demand. At 10 Mt/yr, the 2029 switch from 45Z to 45Q cuts federal value from \$2.0–2.6B to \$0.85B a year.

Where Tallgrass sits



How Tallgrass applies it

- **Own the storage-stage LCA:** one audited data package that all 12 shippers can cite, for a fee or revenue share.
- **Grade tonnes by full LCA:** premium tonnes to strict buyers (Microsoft, Frontier), the rest to spot buyers.
- **Screen new CO₂ sources:** coal earns only 45Q (\$85/t) against capture costs of ~\$60–100/t, so it fits only as volume under a utility take-or-pay contract.

Sources: Trailblazer Low-Carbon Market Strategy; Microsoft 2026 CDR criteria; Puro.earth GSC v.5; 26 USC 45Z and 45Q. Figures marked ~ are estimates.